



# آمار و احتمال مهندسی

## Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

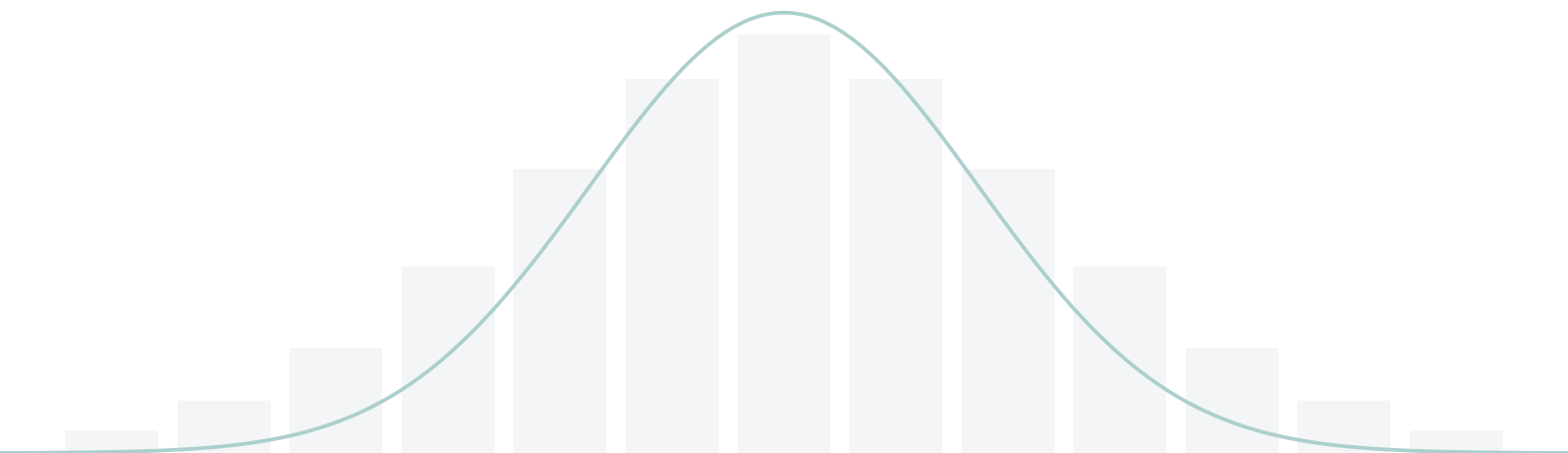
### Module 15

## Hypothesis Testing

آزمون فرض

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### Learning Objectives

After completing this module, students will be able to:

1. Formulate null and alternative hypotheses for engineering problems
2. Identify Type I and Type II errors and their engineering consequences
3. Compute test statistics and p-values
4. Perform Z-tests and t-tests for means
5. Correctly interpret p-values (and avoid common misconceptions)
6. Choose between one-sided and two-sided tests

## 15.1 Introduction: Making Decisions from Data

**Engineering question:** “Is this sensor biased?” “Did the process change?” “Does the new algorithm perform better?”

We need a **systematic framework** to answer such questions from data, while controlling the probability of making errors.

## 15.2 Framework

### ■ Null Hypothesis $H_0$

The **default assumption** — typically “nothing changed” or “no effect”:

- ▶  $H_0: \mu = \mu_0$  (sensor is not biased)
- ▶  $H_0: \mu_1 = \mu_2$  (two methods perform equally)
- ▶  $H_0: \sigma^2 = \sigma_0^2$  (variance hasn't changed)

### ■ Alternative Hypothesis $H_1$

What we want to **detect** or prove:

- ▶  $H_1: \mu \neq \mu_0$  (sensor IS biased — two-sided)
- ▶  $H_1: \mu > \mu_0$  (performance improved — one-sided)
- ▶  $H_1: \sigma^2 > \sigma_0^2$  (variance increased)

### ■ Decision

Based on data, either:

- ▶ **Reject  $H_0$**  → evidence supports  $H_1$
- ▶ **Fail to reject  $H_0$**  → insufficient evidence against  $H_0$

⚠ “Fail to reject  $H_0$ ”  $\neq$  “ $H_0$  is true”. Absence of evidence  $\neq$  evidence of absence.

## 15.3 Types of Errors

|                                 | $H_0$ True (reality)      | $H_1$ True (reality)          |
|---------------------------------|---------------------------|-------------------------------|
| Reject $H_0$ (decision)         | Type I Error ( $\alpha$ ) | Correct! (Power = $1-\beta$ ) |
| Fail to reject $H_0$ (decision) | Correct!                  | Type II Error ( $\beta$ )     |

### ■ Type I Error (False Alarm) — probability $\alpha$

Rejecting  $H_0$  when it's actually true. "Crying wolf."

Engineering analogy: Declaring a process has changed when it hasn't → unnecessary recalibration.

### ■ Type II Error (Missed Detection) — probability $\beta$

Failing to reject  $H_0$  when  $H_1$  is actually true. "Missing the signal."

Engineering analogy: Failing to detect a sensor bias → biased measurements continue.

### ■ Engineering Analogy: Radar Detection

| Statistical Term          | Radar Equivalent         |
|---------------------------|--------------------------|
| $H_0$ : no target         | No aircraft present      |
| $H_1$ : target present    | Aircraft present         |
| Type I error ( $\alpha$ ) | False alarm              |
| Type II error ( $\beta$ ) | Missed detection         |
| Power ( $1-\beta$ )       | Probability of detection |

### ■ The Tradeoff

Reducing  $\alpha$  (fewer false alarms) → increases  $\beta$  (more missed detections), and vice versa.

Fix  $\alpha$  (significance level), then maximize power =  $1 - \beta$ .

## 15.4 Significance Level $\alpha$

The **maximum acceptable Type I error rate**, chosen BEFORE looking at data.

Common choices:  $\alpha = 0.05, 0.01, 0.10$

**Choosing  $\alpha$ :** Consider the cost of a false alarm:

- ▶  $\alpha = 0.01$ : Very conservative (e.g., medical device safety)
- ▶  $\alpha = 0.05$ : Standard in most engineering/science
- ▶  $\alpha = 0.10$ : More tolerant (exploratory analysis)

## 15.5 Power of a Test

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 | H_1 \text{ is true})$$

Power depends on:

1. **Effect size:** Larger true difference  $\rightarrow$  easier to detect  $\rightarrow$  more power
2. **Sample size  $n$ :** More data  $\rightarrow$  more power
3. **Significance level  $\alpha$ :** Larger  $\alpha \rightarrow$  more power (but more false alarms)
4. **Variability  $\sigma$ :** Less noise  $\rightarrow$  more power

## 15.6 The p-value

### Definition

The **p-value** is the probability of obtaining a test statistic **as extreme or more extreme** than the observed value, **assuming  $H_0$  is true**.

$$p\text{-value} = P(\text{data this extreme or more} | H_0 \text{ true})$$

### ■ Decision Rule

- ▶ If  $p\text{-value} < \alpha \rightarrow$  Reject  $H_0$
- ▶ If  $p\text{-value} \geq \alpha \rightarrow$  Fail to reject  $H_0$

### ■ ✓ CORRECT Interpretation

“The p-value is the probability of seeing data this extreme (or more) if  $H_0$  were true. A small p-value means the data is unlikely under  $H_0$ .”

### ■ ✗ WRONG Interpretation

“The p-value is the probability that  $H_0$  is true.”

**Why wrong:**  $H_0$  is either true or false (fixed). The p-value says nothing about  $P(H_0 \text{ true})$ . It measures how surprising the data is under  $H_0$ .

### ■ Calibration

| p-value        | Evidence against $H_0$ |
|----------------|------------------------|
| $> 0.10$       | Weak or none           |
| $0.05 - 0.10$  | Marginal               |
| $0.01 - 0.05$  | Moderate               |
| $0.001 - 0.01$ | Strong                 |

| p-value | Evidence against $H_0$ |
|---------|------------------------|
| < 0.001 | Very strong            |

## 15.7 One-Sided vs. Two-Sided Tests

### ■ Two-Sided ( $H_1: \mu \neq \mu_0$ )

Used when deviation in **either** direction is important.

- ▶ Reject if  $|T| > t_{\alpha/2, n-1}$
- ▶ p-value =  $2 \cdot P(T > |t_{\text{obs}}|)$

### ■ One-Sided ( $H_1: \mu > \mu_0$ or $H_1: \mu < \mu_0$ )

Used when only one direction of deviation matters.

- ▶  $H_1: \mu > \mu_0$ : Reject if  $T > t_{\alpha, n-1}$
- ▶  $H_1: \mu < \mu_0$ : Reject if  $T < -t_{\alpha, n-1}$

### ■ When to Use Which

| Scenario                                 | Test Type                                                 |
|------------------------------------------|-----------------------------------------------------------|
| Is the sensor biased (either direction)? | Two-sided                                                 |
| Does the new algorithm IMPROVE BER?      | One-sided ( $H_1: \text{BER} < \text{BER}_{\text{old}}$ ) |
| Has noise power INCREASED?               | One-sided ( $H_1: \sigma^2 > \sigma_0^2$ )                |
| Did the process mean CHANGE?             | Two-sided                                                 |

## 15.8 Tests for Means

### ■ One-Sample Z-Test ( $\sigma$ known)

Test:  $H_0: \mu = \mu_0$

Test statistic:  $Z = (\bar{X} - \mu_0)/(\sigma/\sqrt{n})$

Under  $H_0$ :  $Z \sim N(0,1)$

### ■ One-Sample t-Test ( $\sigma$ unknown)

Test:  $H_0: \mu = \mu_0$

Test statistic:  $T = (\bar{X} - \mu_0)/(S/\sqrt{n})$

Under  $H_0$ :  $T \sim t(n-1)$

### ■ Two-Sample t-Test (comparing two means)

Test:  $H_0: \mu_1 = \mu_2$

Test statistic:  $T = (\bar{X}_1 - \bar{X}_2) / \sqrt{(S_1^2/n_1 + S_2^2/n_2)}$

Approximate df by Welch-Satterthwaite formula.

## 15.9 Test for Variance (Chi-Square Test)

### ■ One-Sample Test

$H_0: \sigma^2 = \sigma_0^2$

Test statistic:  $\chi^2 = (n-1)S^2/\sigma_0^2$

Under  $H_0: \chi^2 \sim \chi^2(n-1)$

### ■ Decision

- ▶  $H_1: \sigma^2 > \sigma_0^2$ : Reject if  $\chi^2 > \chi^2_{\{\alpha, n-1\}}$
- ▶  $H_1: \sigma^2 \neq \sigma_0^2$ : Reject if  $\chi^2 < \chi^2_{\{1-\alpha/2, n-1\}}$  or  $\chi^2 > \chi^2_{\{\alpha/2, n-1\}}$

## 15.10 Step-by-Step Hypothesis Testing Procedure

1. **State hypotheses:**  $H_0$  and  $H_1$
2. **Choose significance level:**  $\alpha$  (e.g., 0.05)
3. **Select test statistic:** Z, t, or  $\chi^2$  depending on scenario
4. **Determine critical value(s)** or compute p-value
5. **Compute test statistic** from data
6. **Make decision:** Reject  $H_0$  if test statistic in critical region (or  $p < \alpha$ )
7. **State conclusion** in engineering terms

## 15.11 Engineering Examples

### Example 1: Is a Sensor Biased?

A sensor should read 0V with no input. 20 measurements give  $\bar{X} = 0.012\text{V}$ ,  $S = 0.03\text{V}$ .

$H_0: \mu = 0$  (no bias),  $H_1: \mu \neq 0$  (biased),  $\alpha = 0.05$

$T = (0.012 - 0)/(0.03/\sqrt{20}) = 0.012/0.00671 = 1.789$

$t_{\{0.025, 19\}} = 2.093$ . Since  $|T| = 1.789 < 2.093 \rightarrow$  **Fail to reject  $H_0$ .**

p-value =  $2 \cdot P(t(19) > 1.789) \approx 0.090$ . Not significant at 5% level.

Conclusion: Insufficient evidence to conclude the sensor is biased.

**Example 2: Did Manufacturing Process Change?**

Before:  $\mu_0 = 100\Omega$  (established). After modification:  $n = 30$ ,  $\bar{X} = 101.2\Omega$ ,  $S = 3.5\Omega$ .

$H_0: \mu = 100$ ,  $H_1: \mu \neq 100$ ,  $\alpha = 0.05$

$T = (101.2 - 100)/(3.5/\sqrt{30}) = 1.2/0.639 = 1.878$

$t_{\{0.025, 29\}} = 2.045$ .  $|T| = 1.878 < 2.045 \rightarrow$  **Fail to reject  $H_0$ .**

**Example 3: Does New Algorithm Improve BER?**

Old algorithm:  $BER_0 = 10^{-3}$ . New algorithm tested:  $n = 50000$  bits, 38 errors.

$BER_{\text{new}} = 38/50000 = 7.6 \times 10^{-4}$

$H_0: p = 0.001$ ,  $H_1: p < 0.001$  (improvement),  $\alpha = 0.05$

$Z = (\hat{p} - p_0)/\sqrt{(p_0(1-p_0)/n)} = (0.00076 - 0.001)/\sqrt{(0.001 \times 0.999/50000)} = -0.00024/0.000141 = -1.70$

p-value =  $P(Z < -1.70) = 0.0446 < 0.05 \rightarrow$  **Reject  $H_0$ .**

Conclusion: Evidence supports that the new algorithm has lower BER.

**Example 4: Has Noise Power Increased?**

Historical:  $\sigma_0^2 = 0.04 \text{ V}^2$ . New measurements:  $n = 25$ ,  $S^2 = 0.058 \text{ V}^2$ .

$H_0: \sigma^2 = 0.04$ ,  $H_1: \sigma^2 > 0.04$ ,  $\alpha = 0.05$

$\chi^2 = (24)(0.058)/0.04 = 34.8$

$\chi^2_{\{0.05, 24\}} = 36.415$ . Since  $34.8 < 36.415 \rightarrow$  **Fail to reject  $H_0$ .**

**15.12 MATLAB Examples****Example 1: One-Sample t-test**

```
%% Is the sensor biased? One-sample t-test
data = [0.012, -0.005, 0.031, 0.008, -0.012, 0.022, 0.015, ...
        0.003, 0.028, -0.001, 0.018, 0.009, 0.025, -0.008, ...
        0.014, 0.007, 0.020, 0.011, -0.003, 0.019];

mu_0 = 0; % Hypothesized mean (no bias)
[h, p_value, ci, stats] = ttest(data, mu_0);

fprintf('Test result: H = %d (1=reject H_0)\n', h);
fprintf('p-value = %.4f\n', p_value);
fprintf('t-statistic = %.3f, df = %d\n', stats.tstat, stats.df);
fprintf('95% CI for μ: [%.4f, %.4f]\n', ci);
```

MATLAB

**Example 2: Two-Sample t-test**

```

%% Compare two algorithms: is there a significant difference?
BER_algo1 = [0.0012, 0.0008, 0.0015, 0.0010, 0.0013, ...
             0.0011, 0.0009, 0.0014, 0.0012, 0.0010];
BER_algo2 = [0.0007, 0.0009, 0.0006, 0.0008, 0.0005, ...
             0.0008, 0.0007, 0.0006, 0.0009, 0.0007];

[h, p_value, ci, stats] = ttest2(BER_algo1, BER_algo2);
fprintf('Two-sample t-test:\n');
fprintf(' H = %d (1=reject H0: means are equal)\n', h);
fprintf(' p-value = %.4e\n', p_value);
fprintf(' t-stat = %.3f, df = %.1f\n', stats.tstat, stats.df);
fprintf(' Mean diff: %.4f\n', mean(BER_algo1) - mean(BER_algo2));

```

MATLAB

**Example 3: Power Analysis**

```

%% Power: How likely are we to detect a true difference?
mu_0 = 0;           % Null hypothesis mean
mu_true = 0.015;    % True mean (sensor has bias of 15mV)
sigma = 0.03;       % Known std dev
alpha = 0.05;

n_values = 5:5:100;
power = zeros(size(n_values));

for i = 1:length(n_values)
    n = n_values(i);
    % Critical value for two-sided test
    z_crit = norminv(1-alpha/2);
    % Power = P(reject | mu_true)
    % Reject when |Z| > z_crit, where Z ~ N((mu_true-mu_0)/(sigma/sqrt(n)), 1)
    ncp = (mu_true - mu_0) / (sigma/sqrt(n)); % Non-centrality parameter
    power(i) = 1 - normcdf(z_crit - ncp) + normcdf(-z_crit - ncp);
end

figure;
plot(n_values, power, 'b-o', 'LineWidth', 2);
xlabel('Sample Size n'); ylabel('Power (1 - \beta)');
title(sprintf('Power vs Sample Size (true bias = %.0f mV)', mu_true*1000));
grid on;
yline(0.8, 'r--', 'Desired Power = 0.8');
fprintf('Need n \u2248 %d for 80%% power\n', n_values(find(power >= 0.8, 1)));

```

MATLAB

**15.13 Practice Problems**

**Problem 1**

A manufacturing spec requires mean resistance =  $100\Omega$ . A sample of 16 resistors gives  $\bar{X} = 101.5\Omega$ ,  $S = 3\Omega$ . At  $\alpha = 0.05$ , is there evidence the process has drifted?

**Solution**  $H_0: \mu=100$ ,  $H_1: \mu \neq 100$ .  $T = (101.5-100)/(3/4) = 2.0$ .  $t_{0.025,15} = 2.131$ .  $|T|=2.0 < 2.131 \rightarrow$  Fail to reject.  $p\text{-value} \approx 0.064 > 0.05$ .

**Problem 2**

Average packet delay was 5ms. After network upgrade, 40 measurements give  $\bar{X} = 4.6\text{ms}$ ,  $S = 1.2\text{ms}$ . Test if delay decreased at  $\alpha = 0.01$ .

**Solution**  $H_0: \mu=5$ ,  $H_1: \mu < 5$ .  $T = (4.6-5)/(1.2/\sqrt{40}) = -0.4/0.190 = -2.11$ .  $t_{0.01,39} \approx -2.426$ .  $T = -2.11 > -2.426 \rightarrow$  Fail to reject at 1% level. (Would reject at 5% since  $t_{0.05,39} \approx -1.685$ .)

**Problem 3**

Explain why “ $p = 0.04$  means there is only a 4% chance  $H_0$  is true” is WRONG.

**Solution** The p-value is  $P(\text{data this extreme} \mid H_0 \text{ true})$ , NOT  $P(H_0 \text{ true} \mid \text{data})$ . It conditions on  $H_0$  being true and asks about data extremity. To get  $P(H_0 \text{ true} \mid \text{data})$  would require Bayesian analysis with a prior. The p-value is a property of the data-generating process under  $H_0$ , not a posterior probability about  $H_0$ .

*Next Module: Module 16 — Engineering Statistical Applications*